



Azure Cosmos DB is a fully managed NoSQL database service for building high-performance, scalable, and globally distributed applications. It offers low-latency data access with comprehensive SLAs covering availability, throughput, latency, and consistency.

## Key Features:

- **Global Distribution:** Automatically replicate data across multiple regions, providing high availability and low latency for users globally with just a few clicks.
- **Multi-Model Support:** Supports multiple data models, including document (JSON), key-value, graph (Gremlin), and column-family (Cassandra), within the same database service.
- **Guaranteed Performance:** Offers SLA-backed guarantees for single-digit millisecond read and write latencies and 99.999% availability.
- **Flexible Consistency Models:** Provides five consistency levels - Strong, Bounded Staleness, Session, Consistent Prefix, and Eventual, allowing you to fine-tune performance and consistency trade-offs.
- **Scalability & Throughput:** Offers automatic and elastic scaling with provisioned throughput and serverless options, enabling you to scale as your application needs grow.

## Architecture Components:

- **Containers:** Logical units within a Cosmos DB database that hold your items (documents, key-value pairs, graph nodes/edges) and are automatically partitioned for scalability.
- **Partitioning:** Data is automatically partitioned based on the partition key, allowing for unlimited scale and efficient querying.
- **Global Distribution:** Multi-master replication and automatic failover ensure your data is available and consistent across all configured regions.
- **Request Units (RUs):** Performance is measured in Request Units (RUs), representing the throughput needed for operations. RUs can be provisioned or dynamically scaled based on usage patterns.

What is Azure Data Factory?

- Azure Data Factory (ADF) is a fully managed cloud-based data integration service that enables the creation, scheduling, and management of data pipelines.
- It is used for Extract, Transform, Load (ETL) and Extract, Load, Transform (ELT) processes, allowing data movement and transformation across different services and environments.

## Key Features:

- **Data Integration:** Connect and integrate data from over 90+ supported data sources.
- **Data Transformation:** Perform data transformation at scale using Mapping Data Flows, SQL, and Python.
- **Orchestration:** Automate and manage complex data workflows with triggers, control flows, and scheduling.
- **Scalability:** Scalable processing powered by Azure's serverless infrastructure.

## Core Components of Azure Data Factory

- **Pipelines:** A logical grouping of activities that together perform a task. A pipeline is the foundation of an ADF workflow.
- **Activities:** Tasks inside a pipeline, such as Copy Data, Data Flow, Execute SQL, etc.
- **Data Flows:** Visual, code-free data transformations using Apache Spark. Used for complex data mapping, filtering, joining, and aggregations.
- **Datasets:** Representations of data within ADF. Datasets point to data stored in data sources (e.g., Azure Blob Storage, SQL Server).
- **Linked Services:** Connections to data sources or compute services. For example, linking to Azure Blob Storage or an Azure SQL Database.
- **Triggers:** Mechanisms to start pipelines. Can be time-based (schedule) or event-based (file arrival).

## Common Use Cases for Azure Data Factory

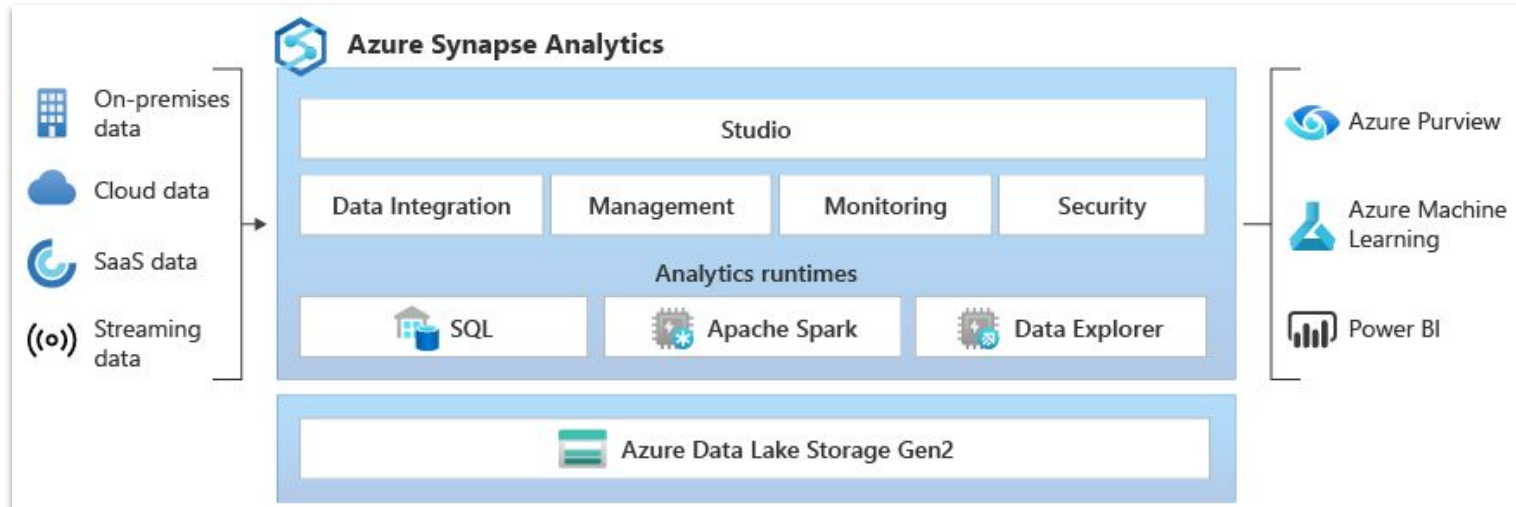
- **Data Migration:** Seamlessly migrate data between different environments (e.g., on-premises to cloud) or between different cloud services.
- **Data Warehousing:** Load and transform large volumes of data from diverse sources into a data warehouse, enabling analytics and reporting.
- **Data Orchestration and Automation:** Orchestrate complex ETL workflows that require multiple steps and involve dependencies between activities.
- **Big Data Integration:** Integrate and process big data from data lakes and IoT devices with support for Hadoop, Spark, and Databricks.

# Azure Synapse Analytics

Azure Synapse Analytics is an integrated analytics service that brings together big data and data warehousing. It offers end-to-end solutions for data integration, data warehousing, and big data analytics.

## Key Features:

- Unified analytics experience
- Serverless and provisioned compute options
- Seamless integration with Azure Data Lake
- Powerful data integration and orchestration tools
- Integrated machine learning capabilities



## Core Capabilities of Azure Synapse Analytics

- **Data Integration:** Unified ETL/ELT process using Synapse Pipelines
- **Data Warehousing:** Traditional data warehousing capabilities
- **Big Data Analytics:** Support for Apache Spark and built-in connectors
- **Real-Time Analytics:** Integration with Azure Stream Analytics
- **Machine Learning Integration:** Embedded ML models using Azure Machine Learning and Synapse Studio

## Key Terminologies in Azure Synapse Analytics

- **Workspace:** Centralized development environment to build analytics solutions
- **SQL Pool:** Dedicated SQL resources for data warehousing workloads
- **Spark Pool:** Apache Spark integration for big data processing
- **Serverless SQL Pool:** Querying data without the need to provision resources
- **Data Lake Storage:** Primary storage for Synapse, integrating with ADLS Gen2
- **Synapse Studio:** Web-based interface for unified analytics, development, and management

## Key Components of Azure Synapse Analytics

- **Synapse Pipelines:** ETL/ELT orchestration for data integration and transformation
- **Data Flow:** Visual data transformation capabilities for both ETL and ELT scenarios
- **SQL Analytics:** Support for T-SQL queries on structured and unstructured data
- **Apache Spark:** Big data processing engine with built-in support for Python, Scala, Java, and R